Ligand-based filtering using Scipion-chem

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Ligand-based filters

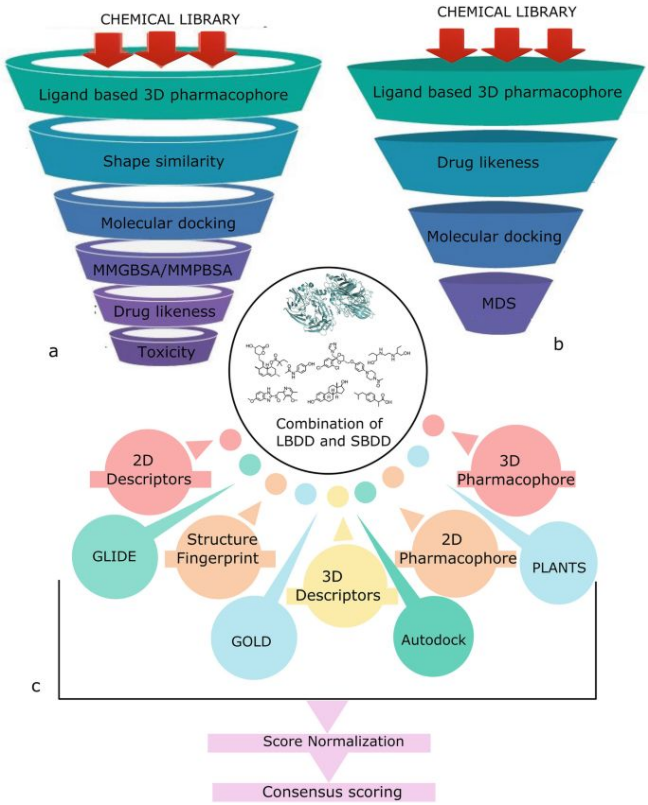
- **Reduce** a compound library before more computationally expensive screening steps
- **Property-based filtering:** Select or remove compounds according to their physicochemical properties and potential liabilities
- **Similarity-based filtering:** Prioritize compounds based on their similarity to known active ligands

Property-based filtering – What are the characteristics of the molecule?

- Molecular properties, drug-likeness rules, predicted ADMET profiles and structural alerts
- Helps identify compounds with suitable properties and flag potential liabilities

Similarity-based filtering – How similar is the molecule to a known active ligand?

- Molecular fingerprints, 3D shape and pharmacophore features
- Helps identify compounds that share chemical features or interaction patterns with known active ligands

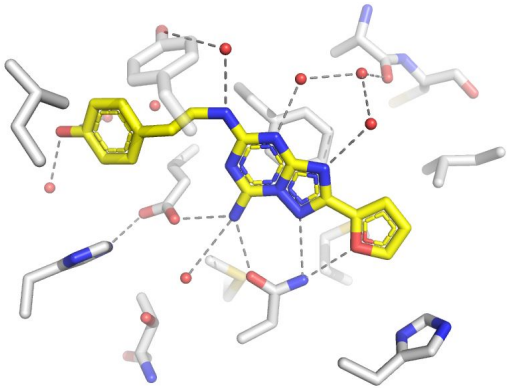


Property-based filtering

- **Goal:** Filter compounds based on their **physicochemical properties** and potential liabilities

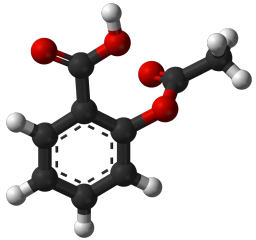
- **Physicochemical properties**
  - Molecular weight and size: Influence membrane permeability and binding-site accessibility
  - Number of atoms and atom types (B, N, O ...): Determine molecular composition and influence chemical properties
  - Number of rings and rotatable bonds: Describe molecular rigidity and flexibility
  - Lipophilicity (LogP): Influences water solubility and membrane permeability

- **Drug-likeness rules**
  - Lipinski rule of 5 (favourable oral absorption)
  - Rule of 3 (fragment-based drug discovery)



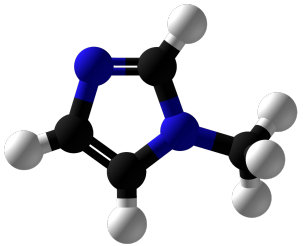
- Mol. weight <=500 Da
- HBD <= 5
- HBA <= 10
- LogP <= 5

- Mol. weight <=300 Da
- HBD <= 3
- HBA <= 3
- LogP <= 3



Aspirin

Methylimidazole

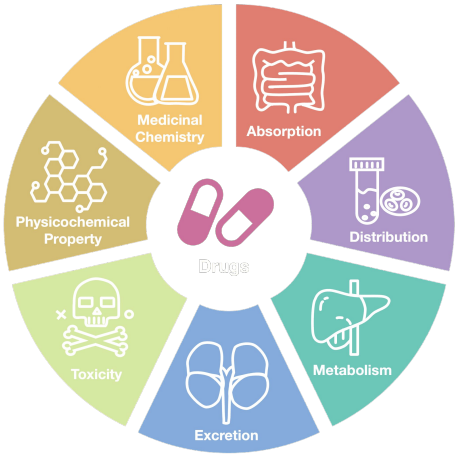


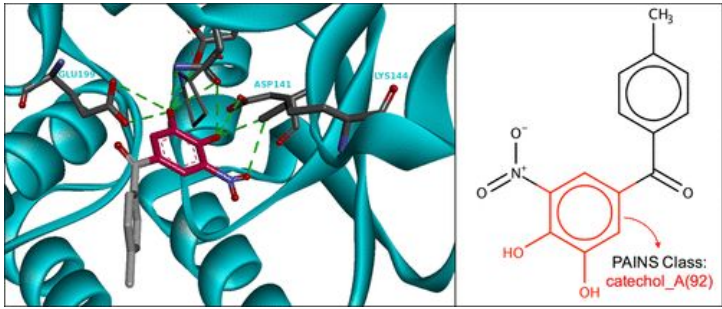
Property-based filtering

- **Goal:** Filter compounds based on their **physicochemical properties** and potential liabilities

- **ADMET predictions**
  - **A**bsorption, **D**istribution, **M**etabolism, **E**xcretion and **T**oxicity
  - Estimate selected pharmacokinetic or toxicity-related properties
  - Help prioritize compounds and flag potential development risks
  - Predictions depend on the model, its training data and the property being evaluated

- **Structural alerts: PAINS** (*Pan-Assay Interference Compounds*)
  - Identify chemical substructures associated with potential assay interference or misleading activity results.
  - Flag compounds for further inspection or experimental validation.

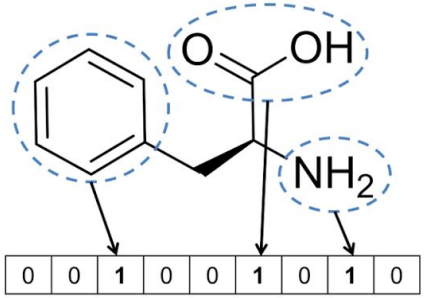




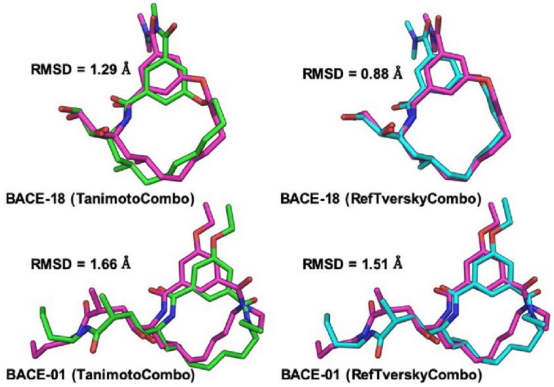
Similarity-based filtering

- **Goal: Identify** compounds **similar to a known active** ligand to **prioritize** candidates

- **Fingerprint similarity**
  - Encode features → substructures, atom environments or chemical patterns into a numerical representation
  - The fingerprints of two molecules can be compared using similarity metrics such as the **Tanimoto coefficient** or **Dice coefficient**
  - Higher similarity = greater overlap between the features represented by the selected fingerprints



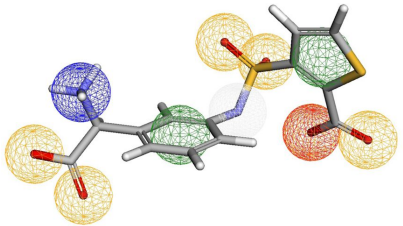
- **Shape similarity**
  - Compares the spatial shapes of molecules
  - Shape-overlap metrics estimate how well the three-dimensional structures match
  - Can identify molecules with similar shapes even when their 2D chemical structures differ



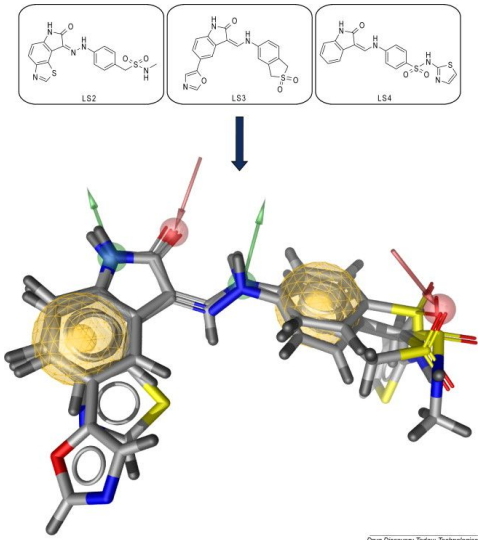
*<u>Key idea</u>: Similarity can **help identify** promising **candidates** but it does **not guarantee biological activity***

Similarity-based filtering

- **Goal: Identify** compounds **similar to a known active** ligand to **prioritize** candidates



- **Pharmacophore similarity**
  - Spatial arrangement of features required or favorable for a molecule to interact with a biological target
    - Hydrogen-bond donors and acceptors
    - Hydrophobic regions
    - Aromatic rings
    - Positively or negatively charged groups
  - Can identify compounds with different chemical scaffolds but similar interaction features
  - Can help prioritize candidates when the known binding mode of an active ligand is available or can be modelled



*<u>Key idea</u>: **DIfferent molecules** can share a **similar pharmacophore** and potentially **interact** with the same **binding site***